

# Firm Learning and Log-Normal Distribution: Evidence from Foreign Direct Investment in Viet Nam

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## Abstract

This paper studies the potential productivity spillover when firms with low productivity are in close proximity to firms with high productivity. Using a firm-level dataset from Viet Nam, I examined the spillover effect of foreign investment firms on domestic Vietnamese firms, which are relatively smaller and have lower value added per person. The exposure is measured in CES form, weighed by spatial proximity and total factor productivity of revenue TFPR, which resulted in a strong positive correlation. Building on the Hopenhayn (1992) model of firm entry and exit, I modeled an economy with different firm types of different log-normal distributions. In this model, low-productivity type learns from the highest type by exposure, which then yields a unique equilibrium. The model calibration shows the trade-off between the low-productivity type's spillover benefits and its market share loss, suggesting an optimal FDI exposure to maximize domestic production in the case of the Vietnamese market.

## 1 Introduction

Firms improve their productivity through various learning mechanisms, one of which is by exposing to firms with more advanced technologies. The miracle economic growth of Asian countries like Japan and South Korea during the nineteenth and twentieth centuries is an example of this process. Their development is frequently attributed to the knowledge gained from interactions with more productive Western enterprises, which typically occurred through foreign investment and direct technology transfers (Cai, 2008). During periods of highest foreign direct investment, growth rate in these countries exceeded 7%, even reaching 10% for Japan in some years (World Bank, n.d.-a, n.d.-b).

Such direct technology transfers are harder to achieve nowadays due to the fiercer global economic competition and perhaps weaker political incentives. Nonetheless, current developing countries in Asia still attempt to mimic such economic condition in a somewhat similar way: attracting foreign direct investment (FDI) into their countries. Specifically, these countries administered favoring conditions for foreign firms, especially in higher technology fields, to either invest in incumbent domestic firms or set up new plants and employ the domestic

labor force. By doing this, developing governments hope that the presence of foreign firms will increase the domestic stock of human capital, induce some forms of within-firm technology transfer in the case of co-operation, raise local firms' awareness of new technologies, or just simply increase demand for domestic inputs, which in turn could incentivize domestic providers to upgrade (Blomström and Persson, 1983). All these channels are often referred to as “productivity spillover,” potentially affecting the productivity of domestic firms in a positive direction.

However, after decades of FDI attraction, it is still unclear whether the benefits outweigh the costs. Besides the possible benefits, the penetration of foreign firms could also create more competition, which in turn can force domestic firms to exit instead of upgrading themselves. Empirical evidence for productivity spillover is also inconsistent, as reported by Demena and van Bergeijk, 2017, in a literature review of 69 FDI-spillover research covering 31 developing countries during the 1986–2013 period. They examined a total of 1450 spillover estimates, which show a mixture of positive, negative, as well as inconclusive results in similar shares.

Although it is challenging to identify an exogenous change in FDI to get an accurate causal effect, this literature review suggests that FDI might not be as helpful as many developing countries have hoped for. In fact, an increase in foreign presence might even hinder the birth and growth of domestic firms as competition increases. It is also possible that there is no learning for high-skilled labor, as foreign firms might just take advantage of the cheap labor force and train them for low-skilled work. Even with positive productivity spillover, domestic and foreign firms might not incorporate it to their profit-maximization problem, leading to an economy smaller than optimal.

To address these problems and trade-offs, this paper shows a one-country general equilibrium with multiple entrant types. The model is built based on the firm entry and exit model of Hopenhayn, 1992, with some simplifications used in Melitz, 2003, for firm production function. This paper also incorporates productivity spillover and used different distributions for different firm types. The model is calibrated using the Viet Nam Enterprise Census data. Details on data are included in Appendix A.

### ***Why Viet Nam?***

Several factors support the choice of Viet Nam for calibration. First, it is a very young country in terms of economic development, with consistent firm census covering the period before and after its first big wave of FDI in 2005. In particular, Viet Nam started a total liberalization in 1994 after decades of centralized economy, wars, and highly limited trade opportunities due to the US sanction. In 2005, its government formalized laws to attract foreign direct investments (FDI), enforcing income tax cut, import tariff exemptions, internalization

exemption, and rent reduction for foreign investors, especially those investing in low-income areas and within innovative sectors. Its firm census is carried out annually from 2000 until now, covering almost the entire period of liberalization and foreign direct investment.

Secondly, not all countries attract FDI the same way – a country might limit foreign ownership by some specific percentage, requiring some ratio of internalization (forcing foreign firms to use some level of domestic inputs), or putting a ceiling on tax exemptions. Viet Nam is a country where there is not much limit on foreign operation: foreign entrepreneurs are allowed to open companies which has 100% foreign ownership, tax and tariff cuts are unlimited, and there is no internalization quota. This has led to a high volume of FDI over the years, creating several new economic zones, resulting in a large deviation in FDI exposure by location.

Third, the timing and composition of Viet Nam’s growth coincide with the direction of FDI exposure, suggesting some positive spillover. For the past 20 years, Viet Nam’s GDP has steadily grown at around 7 percent annually, and FDI inflow has also been keeping up at around 20 percent of total investment and is equivalent to 6 percent of GDP (General Statistics Office of Vietnam, [n.d.](#), World Bank, [n.d.-b](#)). While the number of FDI-related firms is less than 3 percent of all firms physically present in Viet Nam, they employ around 30 percent of the total labor force as of 2018. Most foreign projects focused on the manufacturing sector, especially in the electronic production and assembly, which are potential sectors for technology spillover. After less than 20 years of growing FDI entrance, Viet Nam’s share of manufacturing in total value-added increased from 19 to 24 percent, while employment share increased from 17 to 33 percent (General Statistics Office of Vietnam, [n.d.](#)).

Previous empirical results in Viet Nam show that there is negative horizontal spillover, positive backward spillover, and negative forward spillover from FDI (Huynh et al., [2021](#), Thang et al., [2016](#), Le and Pomfret, [2011](#)). The negative horizontal spillover can be explained by (1) a lack of knowledge sharing as firms are in direct competition within the same sector, and (2) the presence of foreign firms bidding up input prices and taking up market share. The positive backward spillover can be explained by higher input demand from foreign firms, which then motivates domestic suppliers to upgrade. The negative forward spillover, on the other hand, can be a result of high-productivity FDI firms only exporting and not supplying to domestic firms, while hindering the growth of domestic suppliers. There are also other results showing positive horizontal spillover effects (Hoang and Pham, [2010](#)), which can be a result of same-sector learning.

The structure of this paper is as followed: Section [2](#) summarizes the empirical findings, which motivate the set up of the model. Section [3](#) describe the model that integrates learning with exposure. Section [4](#) describes the method of calibration, as well as calibration results. Finally, the last section summarizes the paper.

## 2 Empirical observations

In this section, I used the Viet Nam Enterprise Survey (VNES) from 2000-2018. This survey is conducted by the Vietnamese government, featuring almost all firms in Viet Nam with several small exceptions (explained in Appendix A.1). The dataset features an increasing number of firms, from around 40,000 firms in 2000 to more than 600,000 firms in 2018. Each firm is classified into a 5-digit sector and one of 13 mutually-exclusive types of ownership. Firms’ end-of-year statements are reported annually, including revenue, profit, cost of goods sold, employment, wage, etc.

### *Classification of firms*

The classification of firm type is crucial in our analysis. Originally, there are 13 types of firms in the survey, which are mutually exclusive as each firm can only choose one type to report. Table 1 shows the original types (right column) and the grouped types used in this paper (left column). In this paper, “domestic firms” or low-productivity firms often refers to the first 3 groups, while “FDI-related firms” or high-productivity firms refers to the last 3 groups. Group 1 and 5 (government-related) are occasionally dropped.

| <b>Groups</b>                        | <b>Original types in the survey</b>                                                                                                                                                             |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Government dominated (no FDI)</i> | Ltd. Co. 1 member, having 100% central state capital<br>Ltd. Co. 1 member, having 100% local state capital<br>Joint stock Co., Ltd. Co., having state capital more than 50%<br>State enterprise |
| <i>Public</i>                        | Collective/Unions collective<br>Joint stock Co., not having state capital<br>Joint stock Co., having state capital not exceeding 50%                                                            |
| <i>Private (no FDI)</i>              | Private enterprise<br>Collective name<br>Private Ltd. Co., Ltd. Co., having state capital not exceeding 50%                                                                                     |
| <i>FDI 100%</i>                      | 100% foreign capital                                                                                                                                                                            |
| <i>FDI &amp; government</i>          | Joint venture between State enterprise & foreign                                                                                                                                                |
| <i>FDI &amp; non-government</i>      | Joint venture between other enterprise & foreign                                                                                                                                                |

Table 1: *Classification of firm types*

*Note:* This table shows the definition of firm types by the author (left column) and by the Viet Nam Enterprise Survey (right side). The survey is in Vietnamese, and the translated type names are provided by the General Statistic Office of Viet Nam.

### *Measuring TFPR*

An important metric used in this section is the measure of productivity, TFPR. To calculate this, I assumed a Cobb-Douglas production function with labor and capital inputs. Two different methods were used. The first one is using separate OLS regressions for different

sectors<sup>1</sup>. The second method is by Olley and Pakes, 1996, with the correction in Akerberg et al., 2015, also run separately for each sector. Both methods give very similar results at the end, but the OLS estimation keeps more observations than the Olley-Pakes. All results in this paper come from the second method, but robustness checks include the same analysis using the first method.

In particular, the production function of firm  $i$  in sector  $s$  at time  $t$  is assumed to be Cobb-Douglas. In logs, they can be written as:

$$\log(\text{valueadded}_{ist}) = \log(\text{TFPR}_{ist}) + \alpha_{ks}k_{ist} + \alpha_{ls}l_{ist}$$

in which  $l_{ist}$  is log of number of employees,  $k_{ist}$  is log of fixed capital, and  $(\alpha_{ls}, \alpha_{ks})$  are their sector-level coefficients. Variables  $(\text{valueadded}_{ist}, \text{TFPR}_{ist})$  stand for value added of each firm and their measured TFPR as the names suggested. All values except number of labor are normalized to 2010 price using the consumer price index. The following subsections show the key observations obtained from the data.

It is also important to note that FDI-related firms are treated as the high productivity type, while the domestic firms (not related to FDI) are treated as low productivity type in our analysis. The reason for this classification is in Section 2.2.

## 2.1 Observation: Domestic firms' average TFPR grows at greater rate following the 2006 FDI liberalization

In Figure 1 below, the orange line shows the average of domestic firms' TFPR (in log) for each year from 2000 to 2010. The black dotted line use 2000-05 data to project growth rates starting in 2006, which is the year of FDI liberalization. We can see that before 2006, the overall trend is almost linear with some slight decrease after 2003. However, post 2006, the real growth is higher than the projected growth, suggesting some benefits from the FDI liberalization on domestic firms.

## 2.2 Observation: Foreign firms have a higher productivity distribution

Before answering whether there is any spillover, we need to see if FDI firms are actually more productive than domestic firms. Figure 2 shows the distribution of revenue and productivity for different firm types. We can see that FDI-related firms (green lines) has a higher productivity and revenue distribution than domestic firms (orange lines). For revenue, the shape of the distributions are almost the same, while for TFPR, FDI firms has a denser right-tail and a higher variance. The characteristics of the TFPR graph are similar to those of the value-added distributions.

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<sup>1</sup>A *year* fixed effect is included in the regression, which is then treated as a part of the TFPR.

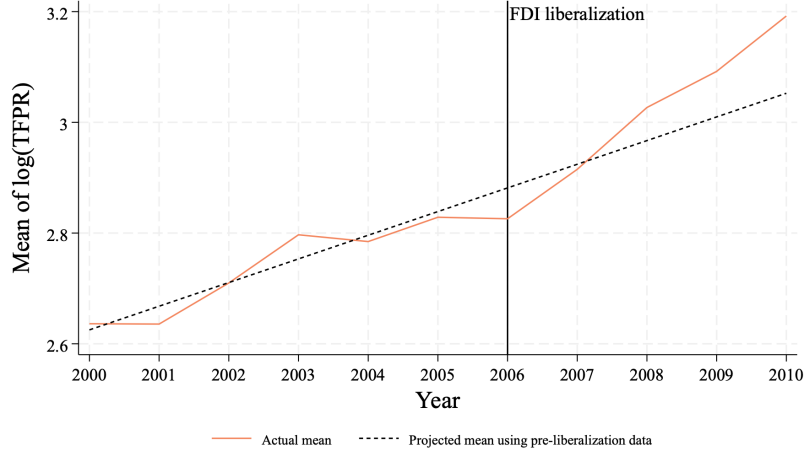


Figure 1: *Trajectory of domestic firms' TFPR over time, 2000-2010*

*Note:* This figure measures the mean of domestic firms' log productivity (measured by TFPR) over time in Viet Nam, using the Viet Nam Enterprise Census and the OLS method of calculating TFPR (in order to keep as many variables as possible). Here, "domestic firms" refers to firms that are not fully or partially owned by FDI.

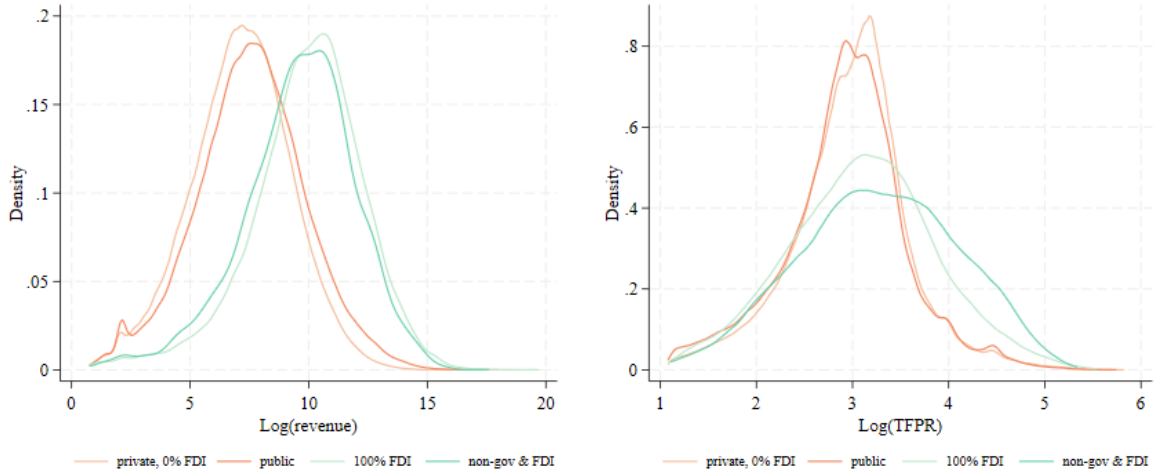


Figure 2: *Comparing different firm types' revenue and TFPR distributions, 2009-2015*

*Note:* This figure compares 4 different firm types in Viet Nam in their revenue and TFPR distribution. Here, I only used manufacturing and service sectors, and the data are combined from 2009 to 2015. Government-related firms are also dropped, as they are not always profit-maximizing firms. This graph uses Stata's `kdensity` command, with Gaussian kernel estimation.

### 2.3 Observation: There is a positive correlation between domestic firms' productivity and FDI exposure.

#### *Measuring total FDI exposure*

A firm's FDI exposure is measured by how far and how productive its surrounding FDI firms are. In the dataset, each firm's area ("province"), sub area ("district"), and sub-sub areas ("ward") are reported, which I used as a proxy of distance between firms and checked for robustness by varying proxy weights. I assumed a CES weight of firms exposure with

elasticity of substitution  $\gamma$ . This does a couple of things: it accounts for the substitutability of knowledge while assuming closer FDI firms brings in higher effect. In our specification, firm size is not accounted for, as in the model later on, higher productivity firms tend to hire more labors, and thus including size and productivity strongly correlate. Finally, FDI exposure of domestic firm  $i$  at time  $t$  is measured as

$$\text{FDI\_exposure}_{i,t} = \left( \sum_{i' \text{ is FDI}} (\text{distance from } i \text{ to } i')^{\frac{1}{\gamma}} \times \text{TFPR}_{i',t}^{\frac{\gamma-1}{\gamma}} \right)^{\frac{\gamma}{\gamma-1}} \quad (1)$$

Since I do not have firm's specific location, distances are approximated by whether two firms share the same region or sub region. In particular, each firm is located by ward, district, and province. Here, I used weight of (1, 0.9, 0.6) if two firms share the same (ward, district, province). <sup>2</sup>.

#### ***Decomposing FDI exposure into backward, forward, and horizontal exposures***

FDI exposure is also decomposed into horizontal, forward, and backward exposures. Here, "horizontal" exposure measures the effect from FDI firms that share the same two-digit sector code with the domestic firm<sup>3</sup>. "Forward" exposure concerns the subset of upstream FDI firms that supplies to the domestic firm, while "backwards" means the subset of downstream FDI firms that buy from the domestic firm. The exact input-output linkage data for each firm is not available; therefore, I used an approximated measure with the Input-Output table from the Asian Development Bank public data library (more details on Appendix A.2).

In particular, for firm  $i$  in sector  $s$ , its backward FDI exposure is very similar to (1) but with an additional backward $_{i,i'}$  weight:

$$\text{backward\_FDI\_exposure}_{i,t} = \left[ \sum_{i' \text{ is FDI}} (\text{backward}_{i,i'}) \times (\text{distance from } i \text{ to } i')^{\frac{1}{\gamma}} \times \text{TFPR}_{i',t}^{\frac{\gamma-1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}}$$

The backward $_{i,i'}$  weight shows the relationship between firm  $i$  (sector  $s$ ) and its buyers  $i'$  (sector  $s'$ ). It is approximated using the Input-Output table at sector level<sup>4</sup>:

$$\begin{aligned} \text{backward}_{i,i'} &\approx \text{Pr}[\text{domestic firm } i \text{ supplies to FDI firm } i' \text{ at year } t] \\ &\approx \text{Sector } s \text{ supply share to sector } s' \text{ at year } t \\ &= \frac{\text{Sector } s \text{ supply to sector } s' \text{ at } t}{\text{Sector } s \text{ total output}} \end{aligned}$$

Forward linkage is defined similarly, but instead of measuring supply of firm  $i$  to FDI firm  $i'$ , it measures the input use firm  $i$  sources from  $i'$ . Horizontal linkage is the most straight-forward,

<sup>2</sup>Table 1. The result is robust for other weights such as (1, 1, 1) and (1, 0.95, 0.8)

<sup>3</sup>There are 35 sectors in total, excluding "export" which is classified as a sector of its own

<sup>4</sup>Note that all forward $_{i,i'}$  will sum to less than 1, as each sector does not only supply to other sectors but also to final consumption.

where we only include FDI firms of the same sector in the calculation.

### *Specification for FDI spillover regressions*

Finally, to measure potential spillover, I used the following identification for domestic firm  $i$  in sector  $s$ , location  $l$ , and time  $t$

$$TFPR_{is,t} = \beta_0 + \beta_1 \times fdi\_exposure_{is,t-1} + \beta_2 \times \mathbf{X}_{is,t} + \beta_3 \times \mathbf{FE}_{is,t} \quad (2)$$

with  $fdi\_exposure_{is,t-1}$  as log of  $TFPR\_exposure_{is,t-1}$  measured in (1) for firm  $i$  of sector  $s$  in year  $t-1$ ;  $\mathbf{X}$  as firm's characteristics such as firm size;  $\mathbf{FE}$  are the province, year, and sector fixed effects. The use of lagged  $fdi\_exposure_{is,t-1}$  variable is aimed at resolving the following problem: there can be some unmeasurable shocks that attract high-productivity FDI entrance while encouraging domestic firms in the same area to improve productivity at the same time. Without the lagged variable, we will have an omitted variable bias.

Regression results for (2) are reported in Table 2. Here, column (1) shows that a domestic firm's past exposure to high productivity FDI firms yields a positive effect on its current firm's productivity. Regression (3) shows that when using a balanced panel, the effects become smaller but still significant. This means that the entry and exit of firms biased FDI exposure, in which new entrance are likely to positively contributing to the average productivity. This is possibly due to the natural improvement of firm entrance overtime, not necessarily due to learning from FDI firms.

|                                               | (1)                    | (2)                    | (3)                    | (4)                    |
|-----------------------------------------------|------------------------|------------------------|------------------------|------------------------|
| Dependent var: Log(TFPR)                      | (full panel)           | (full panel)           | (balanced panel)       | (balanced panel)       |
| Log(FDI exposure)                             | 0.0313***<br>(0.00413) |                        | 0.0292***<br>(0.00508) |                        |
| Log(horizontal FDI exposure)                  |                        | 0.0274***<br>(0.00557) |                        | 0.0294***<br>(0.00436) |
| Log(forward FDI exposure)                     |                        | 0.0187*<br>(0.0100)    |                        | 0.0178**<br>(0.00883)  |
| Log(backward FDI exposure)                    |                        | -0.00151<br>(0.00539)  |                        | -0.00314<br>(0.00354)  |
| R-squared                                     | 0.245                  | 0.249                  | 0.266                  | 0.295                  |
| Observations                                  | 540320                 | 365948                 | 127387                 | 70571                  |
| Standard errors in parentheses                |                        |                        |                        |                        |
| * $p < 0.10$ , ** $p < 0.05$ , *** $p < 0.01$ |                        |                        |                        |                        |

Table 2: *Correlation between a domestic firm's productivity with its past exposure to FDI*

*Note:* This table show the regression results for specification (2). While not shown, all four regressions include firm size, province fixed effect, year fixed effect, and sector fixed effect.

This decomposed effects in columns (2) and (4) show a strong positive horizontal effects, insignificant backward spillover, and significant positive forward exposure effect. This can mean three things. First, having many high-productivity firms in the same two-digit sector nearby helps domestic firms learn. Second, having more foreign suppliers round also improves a do-



mestic firm’s productivity. Third, being a supplier to an FDI firms does not have any spillover.

However, these interpretations on decomposed effects in (2) and (4) need to be taken lightly, due to the imperfect measurement of the decomposed exposures ( $forward_{i,i'}$ ,  $backward_{i,i'}$ ,  $horizontal_{i,i'}$ ). They are simply estimates using sector-level input-output linkage. More importantly, vertical linkages do not include firms of the same sector. For example, in the forward exposure for a firm  $i$  in sector  $s$ , I only included upstream FDI firms *outside* of sector  $s$ . While this helps separating “horizontal” and “forward” effects, it also omits a strong source of forward spillover, as firms often supply to firms of the same two-digit sector, such as a fabric manufacturer supplying to a clothing manufacturer. Including the same sector into the measure of forward exposure, however, generates too little variation between horizontal and forward exposures, as a large fraction of firm supply comes from the same two-digit sector.

The signs on Table 2 contradict the previous results for Viet Nam in the literature. The first reason can be the imperfect measure of decomposed exposure, as explained above. Moreover, in this paper, the FDI exposure is aggregated in CES from instead of a linear form. The linear form does not yield a robust regression result (see ***Robustness and alternative specifications*** below for more details), which is why I did not use it. Finally, papers on Viet Nam often used different samples (even when using the same Viet Nam Enterprise Census data) when it comes to choosing the years, provincial areas, and sector levels.

### ***Robustness and alternative specifications***

The above result is robust under two different methods of TFPR calculation (either OLS regression or Olley and Pakes, 1996, with Akerberg et al., 2015, correction), when using different outlier cutoff values, when changing distance weights, and when changing cluster choice. In these cases, there is no noticeable change in Figures 1, 2, 3, or 4. For Table 2, while the coefficients change, their relative ratios remain at a similar level, while the coefficient signs do not change.

However, the above analysis is not robust when using a linear aggregator of FDI exposure. Specifically, when we use a linear form instead of a CES form in (1), the results are either not significant, or switch signs when changing outlier cutoff levels and adding/removing fixed effects. There are a couple of reasons for choosing the CES form in this paper. Not only that it yields a robust result, but it is also consistent with the model setup in Section 3. The CES aggregator also allows for an imperfect substitutability in knowledge learning. For example, a linear aggregator treats two nearby high-productivity firms as perfect substitutes, while the CES exposure enjoys a taste-of-variety in knowledge.

## 2.4 Observation: Firm revenue distribution is approximately log-normal

Figure 3 shows the histogram of all domestic firms' log revenue as well as their Q-Q plot. The histogram is nearly normal, with a normal distribution (black line) for reference. And as  $\log(\text{revenue})$  behave like a normal distribution, this suggests that  $\text{revenue}$  is a log-normal distribution.

The Q-Q plot (left side of Figure 3) further validates the normality of  $\log(\text{revenue})$ . The center of the histogram shows a normal behavior almost perfectly. There is a high deviation at the lower end, which is most likely due to low-productivity firms exiting and entering. The slight deviation at the higher end can be due to a Pareto tail behavior of the top-performing firms, which we will not focus on for the modeling section.

Figure 4 shows the same statistics but for FDI-related firms. While this figure exhibits a normal trend for the middle part of the data, FDI-related firms have a thicker left tail than a normal distribution. Nonetheless, we will still assume a log-normal distribution for FDI-related firms in the model. It is also important to note that not only the distribution of all firms during all years in the sample is approximately log-normal, this result is also robust when testing by each year and each type of firm among the 6 main types.

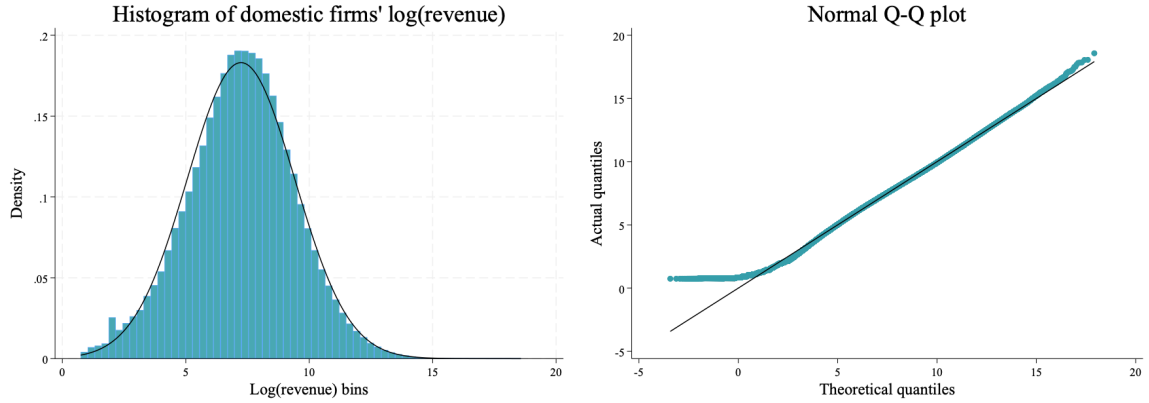


Figure 3: *Testing the log-normality of revenue distribution for domestic firms*

*Note:* The left figure shows the histogram of all domestic manufacturing and service firms in Viet Nam from 2009 to 2015, with a normal distribution line (black line) for reference. The right figure shows the Q-Q plot for this data. Government-related firms are omitted.

## 3 Model

The setup is similar to Hopenhayn, 1992 with some simplifications from Melitz, 2003. The biggest deviation from these two models is the multiple types of entrances that face in different exit rates and fixed costs. Drawing on the previous empirical observations, I also imposed the log-normal distribution on firm productivity draw. We also take on a somewhat naive stance,

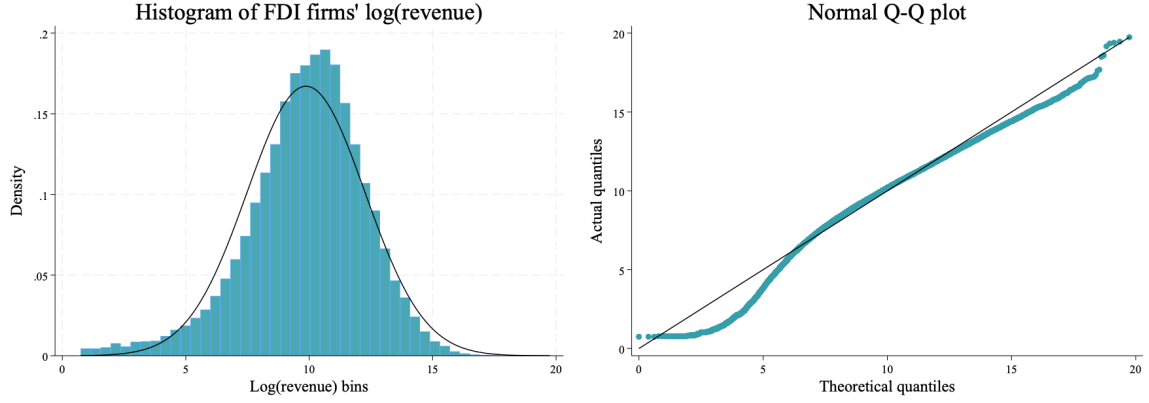


Figure 4: *Testing the log-normality of revenue distribution for FDI-related firms*

*Note:* The left figure shows the histogram of all FDI-related manufacturing and service firms in Viet Nam from 2009 to 2015, with a normal distribution line (black line) for reference. The right figure shows the Q-Q plot for this data.

as suggested by the regressions in Section 2, that there is a positive productivity spillover from high-productivity firms to the low ones.

### 3.1 Household

A representative household has a C.E.S utility function over a continuum of goods indexed by  $\omega$ :

$$Q = \left[ \int_{\omega \in \Omega} q(\omega)^{\frac{\epsilon-1}{\epsilon}} d\omega \right]^{\frac{\epsilon}{\epsilon-1}}$$

Here,  $\Omega$  is the set of all differentiated goods  $\omega$ ;  $\epsilon > 1$  is the elasticity of substitution; and  $q(\omega)$  is the quantity of good  $\omega$  purchased.

Assume further that there are subgroups of goods  $\{\Omega_i\}_{i=1,2,\dots,I}$ , each belongs to a firm types  $i$  (for example, goods produced by foreign firms and goods produced by domestic firms) such that  $\{\Omega_i\}_{i=1,2,\dots,I}$  form a partition of  $\Omega$  and

$$Q_i \equiv \left[ \int_{\omega \in \Omega_i} q(\omega)^{\frac{\epsilon-1}{\epsilon}} d\omega \right]^{\frac{\epsilon}{\epsilon-1}}, \forall i = 1, 2, \dots, I$$

This means total demand can be rewritten as

$$Q = \left( \sum_{i=1}^I Q_i^{\frac{\epsilon-1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon-1}} \quad (3)$$

Given prices  $\{p(\omega)\}_{\omega \in \Omega}$ , household maximizes their utility  $Q$  and set the aggregate price indexes

$$P_i = \left[ \int_{\omega \in \Omega_i} p(\omega)^{1-\epsilon} d\omega \right]^{\frac{1}{1-\epsilon}}, \forall i = 1, 2, \dots, I$$

$$P = \left( \sum_{i=1}^I P_i^{1-\epsilon} \right)^{\frac{1}{1-\epsilon}} \quad (4)$$

The above price indexes are results of the CES demand function. Denoting spending on good  $\omega$  as  $r(\omega) = p(\omega)q(\omega)$ , we also have

$$R_i \equiv \int_{\omega \in \Omega_i} r(\omega) d\omega = P_i Q_i$$

$$R \equiv \sum_{i=1}^I R_i \quad (5)$$

Labor income is derived from its labor  $L$  (supplied inelastically with no unemployment), and wage  $w$  (will be normalized to 1). Total labor income is  $wL$ .

### 3.2 Incumbent firms

There are  $I$  types of firms operating. Each type  $i$  has a mass  $M_i$  and productivity distribution  $\gamma_i(\varphi)$  with  $\varphi \geq 0$ . Each firm produces using only labor  $q(\varphi) = \varphi l$ , given a fixed operating cost  $f_i$  (in unit of labor and is unique to each type  $i$ ), and labor unit cost  $w$ . Each firm's productivity  $\varphi$  is constant overtime.

In this economy, each firm takes quantity demanded as given and set price to maximize profit. Using the variable cost  $\frac{w}{\varphi}$  from the production function, as well as the C.E.S. demand structure with elasticity of substitution  $\epsilon$ , optimal pricing for firms with productivity  $\varphi$  is:

$$p(\varphi) = \frac{\epsilon}{\epsilon - 1} \frac{w}{\varphi} \quad (6)$$

Denote  $\pi_i(\varphi)$ ,  $\Pi_i$ ,  $\Pi$  as profit for firm with productivity  $\varphi$ , total profit for all firms type  $i$ , and total profit in economy. Optimal pricing (4) means

$$\pi_i(\varphi) = \frac{1}{\epsilon} r(\varphi) - w f_i \quad (7)$$

Let firm type  $i$ 's exogenous exit rate be  $\delta_i$ , then a firm type  $i$  and productivity  $\varphi$  has the value function:

$$v_i(\varphi) = \max \left\{ \sum_{t=0}^{+\infty} (1 - \delta_i)^t \pi_i(\varphi), 0 \right\} = \max \left\{ \frac{1}{\delta_i} \pi_i(\varphi), 0 \right\}$$

This value function means that a firm exits if their profit falls below 0. In fact, we can identify a cutoff  $\varphi_i^*$  for productivity such that firm type  $i$  will exit if their productivity falls below  $\varphi_i^*$ . From the C.E.S. demand, it must be true that

$$q(\varphi) = Q P^\epsilon p(\varphi)^{-\epsilon} \quad (8)$$

which means

$$\frac{q(\varphi)}{q(\varphi')} = \left(\frac{\varphi}{\varphi'}\right)^\epsilon \text{ and } \frac{r(\varphi)}{r(\varphi')} = \left(\frac{\varphi}{\varphi'}\right)^{\epsilon-1} \quad (9)$$

for any two firms  $\varphi, \varphi' \in \Omega$ . This and (7) mean  $\lim_{\varphi \rightarrow 0^+} \pi_i(\varphi) = -f_i < 0$  and  $\lim_{\varphi \rightarrow \infty} \pi_i(\varphi) = \infty$ . We also have  $\pi_i$  is continuous and strictly increasing on  $(0, \infty)$  if all firm produces. These conditions ensure that there is a unique cutoff  $\varphi_i^*$  of productivity for each firm type  $i$  to be profitable, such that

$$\pi_i(\varphi_i^*) = 0 \Leftrightarrow \frac{1}{\epsilon} r(\varphi_i^*) - w f_i = 0, \forall i = 1, 2, \dots, I \quad (10)$$

This is the zero-profit cutoff condition (ZPC).

### 3.3 Simplifying the expressions

Similar to Melitz, 2003, the economy can be characterized by representative productivities instead of the whole distributions. In particular, define the average productivity of each firm type as

$$\tilde{\varphi}_i = \left[ \int_{\varphi_i^*}^{\infty} \varphi^{\epsilon-1} \gamma_i(\varphi) d\varphi \right]^{\frac{1}{\epsilon-1}} \quad (11)$$

with  $\gamma_i$  being the distribution of incumbent firms of type  $i$ . Then, we have

$$P_i = M_i^{\frac{1}{1-\epsilon}} p(\tilde{\varphi}_i) \quad (12)$$

$$Q_i = M_i^{\frac{\epsilon}{\epsilon-1}} q(\tilde{\varphi}_i) \quad (13)$$

$$R_i = M_i r(\tilde{\varphi}_i) \quad (14)$$

$$L_{pi} \equiv M_i l_i(\tilde{\varphi}_i)$$

with  $L_{pi}$  being the productive labor use to pay variable cost and fixed production costs for all firms in type  $i$ , and

$$l_i(\varphi) = \frac{q(\varphi)}{\varphi} + f_i \quad (15)$$

for all  $\varphi, i$ .

### 3.4 Firm entrance

Each potential firm knows their type  $i$  but only draw a productivity  $\varphi_i$  after paying a fixed entrance cost  $f_{ei}$  (in unit of labor). Each firm type  $i$  draws from productivity distribution with pdf  $g_i(\varphi)$  and cdf  $G_i(\varphi)$  with domain  $(0, \infty)$ . The mass of entrance for each type is  $M_{ei}$ , which is zero if no firm of type  $i$  enters. In a stationary equilibrium, entry equals exit

$$\delta_i M_i = (1 - G_i(\varphi_i^*)) M_{ei}, \forall i = 1, 2, \dots, I \quad (16)$$

as firms type  $i$  drawing below  $\varphi_i^*$  will immediately exit. The stationary distribution for firms type  $i$  is

$$\gamma_i(\varphi) = \begin{cases} \frac{g_i(\varphi)}{1-G_i(\varphi_i^*)}, & \varphi \geq \varphi_i^* \\ 0, & \text{otherwise} \end{cases} \quad (17)$$

Firms type  $i$  tries to enter if and only if fixed entrance cost does not exceed the lifetime expected profit of its type. Taking into account the life-time multiplier  $\frac{1}{\delta_i}$ , the chance of immediate survival  $1 - G_i(\varphi_i^*)$ , and the current mass  $M_i$  of firms, we can characterize the expected lifetime profit of firm type  $i$  prior to productivity draw as

$$\frac{1}{\delta_i} \frac{1}{M_i} \Pi_i[1 - G_i(\varphi_i^*)] = \frac{1}{\delta_i} \left[ \frac{1}{\epsilon} r_i(\tilde{\varphi}_i) - w f_i \right] [1 - G_i(\varphi_i^*)]$$

In a zero expected profit model, firms' free entry condition (FE) means their expected life-time profit cannot exceed their fixed entry cost. This means

$$M_{ei} \begin{cases} > 0, & \frac{1}{\delta_i} \left[ \frac{1}{\epsilon} r_i(\tilde{\varphi}_i) - w f_i \right] [1 - G_i(\varphi_i^*)] = w f_{ei} \\ = 0, & \frac{1}{\delta_i} \left[ \frac{1}{\epsilon} r_i(\tilde{\varphi}_i) - w f_i \right] [1 - G_i(\varphi_i^*)] < w f_{ei} \end{cases} \quad (18)$$

### 3.5 Market clearing

Labor market clearing means

$$L = \sum_{i=1}^I [M_i l_i(\tilde{\varphi}_i) + M_{ei} f_{ei}]$$

Household spends all income, which means

$$R = wL \quad (19)$$

This final market-clearing condition can be proved using the stationary equilibrium condition, firm's revenue decomposition, and free-entry condition<sup>5</sup>.

### 3.6 Learning

The problem is only meaningful if there are higher productivity types with higher fixed costs (if the higher productivity type is also facing lower fixed costs, then they will be the only survivor, which leads us back to the Melitz case of one-type firm economy).

**Assumption 1.** (Firm distribution) Type  $i$  has a log-normal distribution  $g_i$  parameterized

---

<sup>5</sup>Total revenue of the economy is  $R = \sum_i R_i = \sum_i (\Pi_i + wL_{pi}) = \sum_i (\frac{1}{\epsilon} R_i - wM_i f_i + wL_{pi}) = \sum_i \left( \frac{wM_i \delta_i f_{ei}}{1-G_i(\varphi_i^*)} + wL_{pi} \right) = \sum_i (wM_{ei} f_{ei} + wL_{pi}) = wL$ , with the third equality coming from competitive monopolistic mark-up, fourth equality from (18), fifth equality follows from (16), and final equality from the inelastic supply of labor. This proof is similar to Melitz, 2003, page 1704.

by  $\mu_i$  and  $\sigma_i$  such that

$$g_i(\varphi) = \frac{1}{\varphi\sigma_i\sqrt{2\pi}} \exp\left[-\frac{(\ln \varphi - \mu_i)^2}{2\sigma_i^2}\right]$$

for each  $\varphi \in (0, \infty)$ .

**Definition 1.** (Higher productivity distribution) Type  $i$  is defined as *more productive* than type  $i'$  if  $\mu_i \geq \mu_{i'}$  and

$$1 - G_i(x) \geq 1 - G_{i'}(x)$$

for all  $x \geq e^{\mu_{i'}}$ .

Intuitively, this definition states that for any  $\varphi$  above the median of the lower productivity type, the higher productivity type always have a higher accumulated weight  $1 - G_i(\varphi)$ . While this does not fully rationalize the left tail of each distribution, in the case of log-normal distribution, the right-tail is of much more importance.

**Assumption 2.** (Order of distribution) Firm types are ordered such that  $\mu_i \geq \mu_{i+1}$  and  $\sigma_i \geq \sigma_{i+1}$  for all  $1 \leq i \leq I - 1$ .

This definition is inspired by the real-life distribution in Figure 2, in which the FDI-related type has higher mean and median. Intuitively, this definition means two things. First, types with higher median will always have a higher variation in our setup. Second, type 1 has a higher productivity distribution than type 2, type 2 higher than type 3, and so on. The second result<sup>6</sup> follows from the characteristic of log-normal distribution and from Definition 1.

**Assumption 3.** (Learning) Let  $M = \sum_{i=1,\dots} M_i$  be the total mass of firms. If type 1 is the best type, then all other types learn from the existence of type 1 such that for all  $i \neq 1$ ,

$$\mu_i = \underline{\mu}_i + h_\mu\left(\frac{M_1}{M}, \underline{\mu}_i, \mu_1\right) \quad (20)$$

$$\sigma_i = \underline{\sigma}_i + h_\sigma\left(\frac{M_1}{M}, \underline{\sigma}_i, \sigma_1\right) \quad (21)$$

with  $\underline{\sigma}_1 \leq \sigma_i \leq \sigma_1$ ,  $\underline{\mu}_i \leq \mu_i \leq \mu_1$ ,  $\lim_{\frac{M_1}{M} \rightarrow 1} \mu_i = \mu_1$ ,  $\lim_{\frac{M_1}{M} \rightarrow 1} \sigma_i = \sigma_1$ ,  $\partial h_\mu / \partial \frac{M_1}{M} > 0$ , and  $\partial h_\sigma / \partial \frac{M_1}{M} > 0$ .

Intuitively, these assumptions mean two things

- (i) The more type 1 penetrates the economy, the more low-productivity types learn.

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<sup>6</sup>Proof for "if type 1 and 2 follows Assumption 2, then type 1 is more productive than type 2, as defined in Definition 1":

Since both distributions are log-normal, we need to prove  $1 - G_1 = 1 - \Phi\left(\frac{\ln \varphi - \mu_1}{\sigma_1}\right) \geq 1 - \Phi\left(\frac{\ln \varphi - \mu_2}{\sigma_2}\right) = 1 - G_2$  for all  $\varphi \geq e^{\mu_2}$ . This claim is equivalent to  $\frac{\ln \varphi - \mu_1}{\sigma_1} \leq \frac{\ln \varphi - \mu_2}{\sigma_2}$  (\*) for all  $\ln \varphi \geq \mu_2$ . If the left-hand-side of (\*) is weakly negative, then (\*) is true because its right-hand-side is always weakly positive. If the left-hand-side of (\*) is positive, then (\*) is true due to  $\sigma_1 \geq \sigma_2 > 0$  and  $\mu_1 \geq \mu_2$ .

- (ii) When type 1 expands to fully penetrate the market ( $M_1/M \rightarrow 1$ ), all other types are able to identically mimic type 1.

**Assumption 4.** (Fixed costs) The highest-productivity type (type 1) faces the highest fixed production cost

$$f_1 \geq f_i \text{ for all } i = 2, 3, \dots, I$$

and the highest fixed entrance cost normalized by the exit rate

$$\delta_1 f_{e1} > \delta_i f_{ei} \text{ for all } i = 2, 3, \dots, I \quad (22)$$

Here, fixed entrance cost is normalized by the exogenous exit rate because a potential firm looks at not only the cost to enter but also how risky the market is. I want to impose a condition where type 1 not only faces high fixed cost, but also a market risky enough that prevent them from monopolizing. For example, when fixed cost  $f_{e1}$  is high but type 1 does not have any risk ( $\delta_1 \rightarrow 0$ ), while other types have very high risk, then type 1 will be the only player in the market. More information about this follows in Proposition 1's proof.

### 3.7 Competitive equilibrium and solution

**Definition 2.** (Stationary Competitive Equilibrium) Given the exogenous parameters  $L, \epsilon, \{\delta_i, f_i, f_{ei}\}_I, \mu_1, \sigma_1$ , a normalized wage  $w = 1$ , and learning rules  $h_\mu(\cdot), h_\sigma(\cdot)$ , a competitive equilibrium is defined by  $\{P, Q, R, \{P_i, Q_i, R_i, M_i, M_{ei}, \gamma_i(\cdot), \varphi_i^*, \tilde{\varphi}_i, \mu_i, \sigma_i\}_{i=1,2,\dots,I}, \{p(\varphi), q(\varphi), \{l_i(\varphi)\}_{i=1,2,\dots,I}\}_{\varphi \in (0,\infty)}$  such that:

- (i) Household take prices  $\{p(\varphi)\}_{(0,\infty)}$  and characteristic of firms  $\{\tilde{\varphi}_i, M_i\}_I$  as given, then choose quantities to maximizes their utility such that (3), (5), (8), (13), (12), (14) are satisfied.
- (ii) Incumbent firm of type  $i$  with productivity  $\varphi$  take demand  $\{q(\varphi)\}_{0,\infty}$  and fixed cost  $\{f_i\}_I$  as given, choose price  $p(\varphi)$ , labor use  $l_i(\varphi)$ , and exit rule  $\varphi_i^*$  to maximize profit such that (6), (10), (15) are satisfied. Each type's drawing distribution  $g_i$  satisfies learning rules (20) and (21).
- (iii) Potential firms take fixed cost and profitability of incumbent firms as given, and choose whether to enter or exit such that (18) is satisfied.
- (iv) Market is cleared such that (19) is satisfied.
- (v) The economy is at static equilibrium such that (16) and (17) are satisfied given (11).

The solution for this stationary competitive equilibrium satisfies the following properties:

**Proposition 1.** *If  $M_1 > 0$  and  $f_1 = f_i$  for all  $i$ , then there exists  $i \neq 1$  such that  $M_i > 0$ .*

*Proof.* Appendix B.1. □



Intuitively, this proposition states that the most productive type cannot be the only type existing in the market. This is because, as type 1 gets closer to fully penetrate the market, other types get closer and closer to perfectly mimic type 1's productivity distribution. And since lower-productivity types face weakly lower fixed costs, they will enter and prevent type 1 from becoming the only player in the market.

**Proposition 2.** *Let all types have the same  $\sigma$ . When  $M_1 > 0$ , type 1's mass share  $\frac{M_1}{M}$  and each type's distribution are unique.*

*Proof.* Appendix B.2. □

Note that here, we are not considering the case when type 1 does not exist ( $M_1 = 0$ ). This is because (1) it is not interesting regarding the topic of this paper, and (2) there is not change in the distribution that each low type draws from, and the solution can be found using algorithm in Appendix B.5.

**Proposition 3.** *Let all types have the same  $\sigma$  and  $I = 2$ . When  $M_1 > 0$ , the solution for the stationary competitive equilibrium is unique.*

*Proof.* Appendix B.3. □

We will use this result for calibration, when the two types are domestic and foreign firms.

**Proposition 4.** *Average revenue  $r(\tilde{\varphi}_1)$  of type 1 does not depend on  $\{\mu_i\}_{i=1,2,\dots}$  distribution, but only on exit rate  $\delta_1$ , fixed costs  $f_{e1}, f_1$ , and  $\sigma_1$ .*

*Proof.* Appendix B.4. □

This is an interesting result, which says changing  $\mu_1$  does not change  $r(\tilde{\varphi}_1)$ . In our context, it does not matter who the host country opens door to (either lower  $\mu$  foreign country or higher  $\mu$  foreign country), the average revenue of each foreign firm stays constant. This also explains why multiple FDI countries can enter and survive at the host location despite their technological differences. In the case of Viet Nam, the dominant FDI countries include China, the US, South Korea, and Japan, even though their technological differences are vast.

This is a nice characteristic whose proof depends heavily on the log-normality of the firm distributions. Intuitively, increasing  $\mu_1$  means higher competition within foreign firms. And since the entry decision for type 1 only depends on expected profit of its own type, only foreign firms with better productivity draw choose to stay upon arrival.

Other interesting results related to this stationary competitive equilibrium can be found in Appendix D.

## 4 Model calibration and counterfactuals

In the calibration, we will use a two-type model with log-normal productivity, where the mean and variance of the low type’s growth depends on the penetration of the high type. I used 2001-2006 data to reverse-engineer the parameter of the model, then project the parameters and solve for post-2006 domestic productivity. The parameters used are included the below table, with more details in the next section (parameter estimation strategy).

| Parameters                      |                           | Method                                                                                                                                                         |
|---------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Elasticity of substitution      | $\epsilon = 4$            | Calculate upperbound from aggregate data 2000-05                                                                                                               |
| Fixed costs                     | $f_{i,t}$                 | Range estimated from the reverse-calculation of 2000-08 data                                                                                                   |
| Fixed entrance costs, foreign   | $f_{e1,t}$                | Range estimated from the reverse-calculation from 2000-05 data to estimate range. Final values are chosen within such range to match the real $\mu_{r_f}, t$ . |
| Fixed entrance costs, domestic  | $f_{e2,t}$                | Range estimated from the reverse-calculation from 2000-05 data. Final values are chosen within such range to match the real $\mu_{r_d}, t$ .                   |
| Productivity, foreign           | $\mu_{f,t}, \sigma_{f,t}$ | Reverse-calculation from 2000-05, project trend for 2006-08                                                                                                    |
| Baseline productivity, domestic | $\mu_{i,t}, \sigma_{i,t}$ | Matching for joint moments of domestic firm revenue distribution, zero-profit cutoff, and free-entry condition for 2000-05. Project trend for 2006-08.         |
| Exit rate                       | $\delta_i$                | Real exit rate, 2000-08                                                                                                                                        |

Table 3: *Parameter calibration strategy*

### 4.1 Parameters estimation strategy

I treated each year in the train dataset as a competitive general equilibrium and used structural accounting to characterize the values and movement overtime of the exogenous parameters. In particular, each year’s data observes the entry and exit over time of each firm, the distribution of revenue and labor use  $\{r_i, l_i\}_i$  for all firms in the manufacturing and service sectors, number of firms  $M_i$  for each firm type, and total labor pay  $wL = \sum_i w_i l_i$ . From these, I estimated the elasticity of substitution  $\epsilon$ , exit rate  $\delta_{it}$ , and distribution of productivity  $(\mu_{it}, \sigma_{it})$ . I then checked the model’s accuracy using 2005 data, and ran counterfactual using 2006-2008 data.

In the counterfactual exercise, there will be three relative paths of growth. First, I used the above results from train data to characterize a hypothetical growth path as if there were no 2005 liberalization. Second, I used the learning parameters at firm level (shape of  $h$  functions) extracted from 2009-2015 dataset in Section 2 to estimate the growth with the 2005 liberalization. Third, I compare them with actual growth of domestic productivity.

Appendix B shows how to infer parameters that are not directly observed, such as elasticity of substitution and fixed costs. Assuming that each year is a competitive equilibrium, Appendix B provides proof and method to calculate elasticity of substitution, wage, revenue at special productivity values, fixed costs, exit rate, productivity parameters, and fixed entrance cost.

## 4.2 Calibration results

### Matching of revenue distribution

By construction (table 3), the values of  $\mu_{r_i}$  and  $\sigma_{r_i}$  with match well with the real values, since I chose the best values from the previous section's range to match revenue distribution. For each year, the deviation of  $\mu_{r_i}, \sigma_{r_i}$  from the real values is less than 1%. Figure 5 is an example comparison between actual revenue distribution (blue line) and the calibrated revenue distribution (red line) if there were no exit. While the log-normal parameters  $(\mu_{r_i}, \sigma_{r_i})$  matches

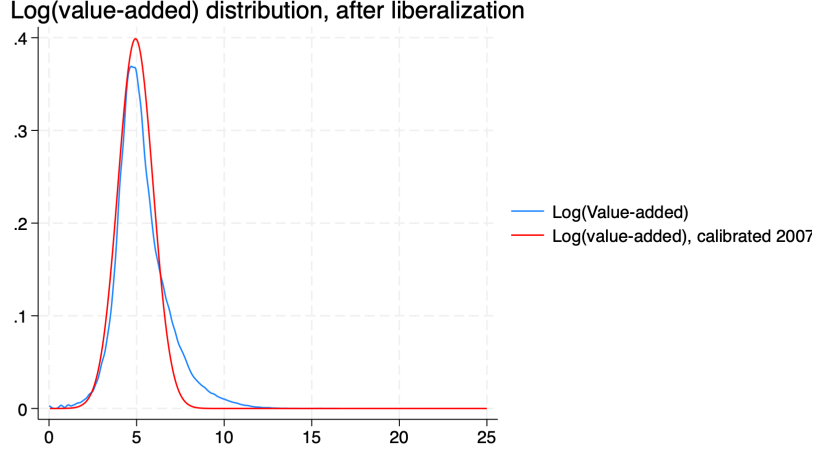


Figure 5: *Matching of revenue distribution*

*Note:* The above calibration (red line) is not showing the cutoff point.

well, the cutoff of productivity does not. This is a common problem with model on firm entry and exit. In the calibrated model, domestic entrants' survival rate is only 10-25%, while it is 22-52% for foreign firms. This means that the revenue distribution for domestic firms should be cutoff at the far right-hand side; while in real life, there is no observed cutoff point.

### Matching of growth in domestic productivity following the 2006 FDI shock

The calibrated model predicts an increase in productivity on top of projected growth (third column of Table 4). This shows the same trend as of real additional growth on top of projected growth (third column of Table 4); however, calibrated variable is more stable and does not explain fully the real growth.

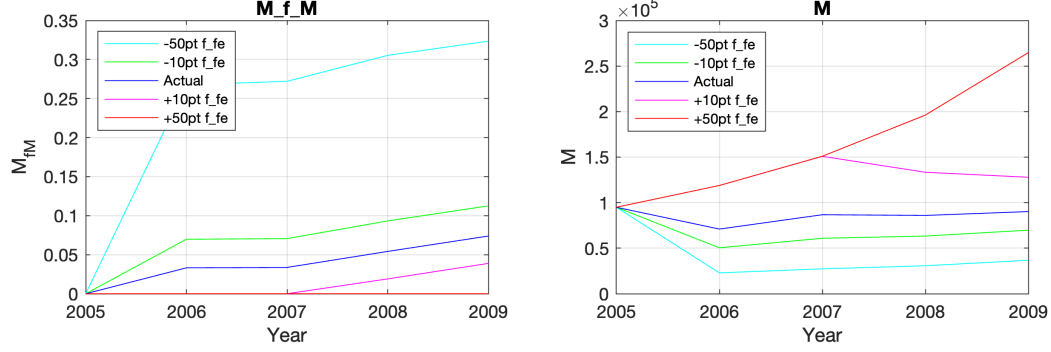
| Year | Actual increase in $\mu_d$ | Calibrated increase in $\mu_d$ |
|------|----------------------------|--------------------------------|
| 2006 | -1.035%                    | 0.093%                         |
| 2007 | 0.296%                     | 0.244%                         |
| 2008 | 2.766%                     | 1.095%                         |
| 2009 | 3.703%                     | 2.354%                         |

Table 4: *Calibrated growth of domestic firms*

### 4.3 Counterfactuals

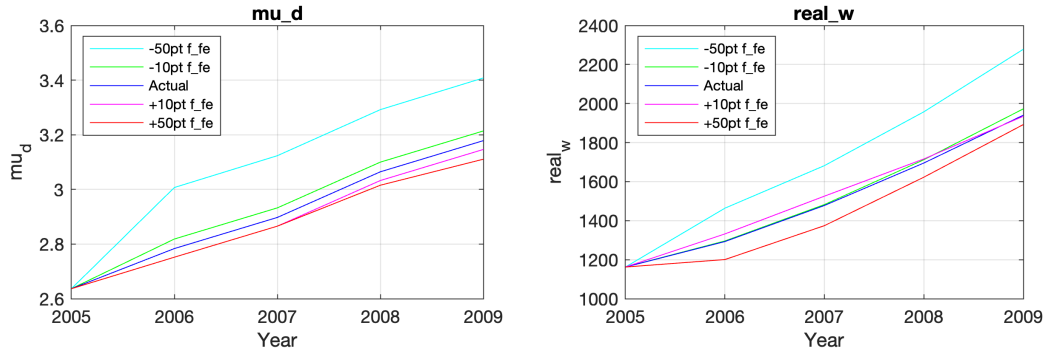
For this section, I ran alternative scenarios where fixed cost for foreign firm to enter ( $f_{e1}$ ) changes by 10 percentage point and 50 percentage point in each direction. I also assumed linear learning, with rates depend on the gap between foreign and domestic parameters.

#### Number of firms



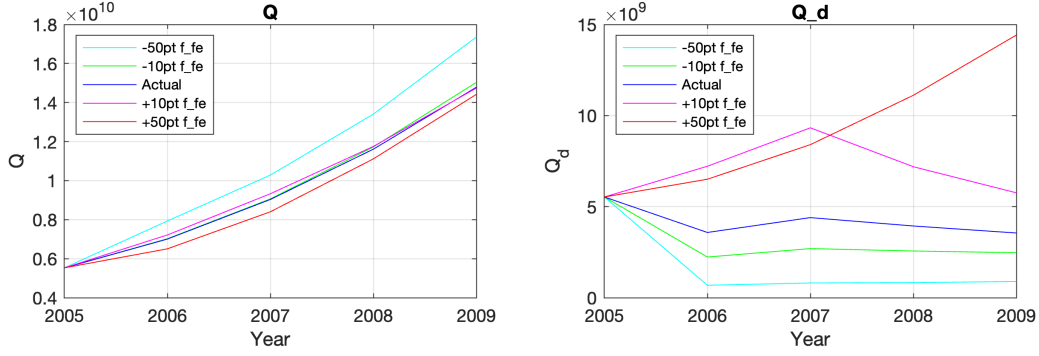
The left graph shows share of foreign firms, while the right graph show the total number of firms. As fixed costs to enter drop, there is a higher share of foreign firms. However, the total number of firms decreases, which can be explained by the increase in overall productivity of firms within the country that increases competition.

#### Domestic productivity and real wage



On the left-hand side, we can see that domestic productivity has a higher  $\mu_d$ . As fixed costs to foreign firms drop, there is a higher share of foreign firms, which then increases  $\mu_d$ . Welfare also increases, as it is the inverse of price aggregate  $P$  which drops due to higher share of high-productivity foreign firms.

#### Total quantity and domestic quantity



Total quantity of production  $Q$  (left graph) increases with lower costs, which can be explained by a higher share of labor moving into productive production. However, total domestic quantity  $Q_d$  drops. This happens despite the increase in  $\mu_d$  that we have seen above. This is because while on average, domestic firms are more productive, fiercer competition makes the number of domestic firms drop, leading to a lower aggregate  $Q_d$ .

## 5 Welfare

There are multiple ways to define welfare in this context. From the consumer point-of-view, welfare only concerns the country's purchasing power (or real wage), which is improved with more variety of goods, as proved in Melitz, 2003. But, domestic government oftentimes care more about the productivity of domestic firms and their share of the market ( $Q_d$ ). Each of these are analyzed in the counterfactual section; however, it is not a comprehensive analysis as it depends on only a single set of parameters.

In this section, we will talk about a more comprehensive welfare analysis for both real wage  $w$  and domestic production  $Q_d$ . The change in policy will be change in fixed costs. In real life, the Vietnamese government often change fixed entry cost  $f_{e1}$  by changing real estate policy, such as establishing foreign economic zones and government offices aiming at reducing knowledge barrier with potential investors, or raising and strengthening favorable land leasing terms. They also change fixed cost to operate  $f_1$  with income tax relief and import tax cut. However, in our analysis, the focus is on fixed entrance cost.

Below is the counterfactual for different policy scenarios regarding fixed entrance cost change. Here, we can see that there is a trade-off between domestic firms' productivity  $\mu_d$  and its

| Policies                             | $\Delta\mu_d$ | $\Delta$ real wage | $\Delta M_f$ share | $\Delta Q_d$ | $\Delta Q$ |
|--------------------------------------|---------------|--------------------|--------------------|--------------|------------|
| ↓ cost to FDI by 50%                 | 8.03%         | 13.22%             | 706.96%            | -80.89%      | 13.22%     |
| ↓ cost to FDI by 10%                 | 1.25%         | 0.23%              | 110.44%            | -37.65%      | 0.23%      |
| ↑ cost to FDI by 10%                 | -1.14%        | 3.02%              | -100.00%           | 101.46%      | 3.02%      |
| elasticity w.r.t. to $\Delta f_{e1}$ | -0.079        | -0.022             | -6.55              | 2.45         | 0.0081     |

Table 5: *Counterfactual: different policy scenarios*

quantity share  $Q_i$ , as seen in the elasticity with respect to  $\delta_{fe_i}$  on the last row of Table 5. Therefore, the government is facing of trade off between learning and domestic share. While having more foreign firms might help domestic firms learn, they are also potentially stealing market share, hindering the growth of domestic entrepreneurs. Nonetheless, we still need a more concrete proposition for this trade-off, instead of using a one-case counterfactual as fact.

## 6 Conclusion

Overall, this paper provided empirical evidences for some positive spillover of productivity from foreign to domestic firms, alongside with the lognormal distribution of firms in Viet Nam. In particular, using the Vietnamese Enterprise Survey (VNES) from 2000 to 2018, I have measured domestic firms' exposure to high-productivity foreign-direct-investment firms (FDI). I tested for each firm's productivity against its past exposure with FDI firms, and found a positive overall correlation, suggesting some productivity exposure between firms of low and high productivity. The exposure is statically significant and positive for horizontal and forward exposure, while not significant for backward spillover.

Observing the log-normal distribution of each firm type using the same Vietnamese dataset, I also constructed a model with CES demand and log-normal productivity distribution. The model is calibrated using the 2006 FDI shock in Viet Nam when it formally recognized FDI firms and provided multiple benefits to foreign investors. The calibrated model explains the growth in domestic productivity in addition to the projected growth path, although it does not fully explain the entire additional growth compared to real data. In addition, the counterfactual results show that while decreasing barrier further for foreign firms lead to the country's higher total production and real wage, it would decrease the share of domestic firms and their total production quantity.

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